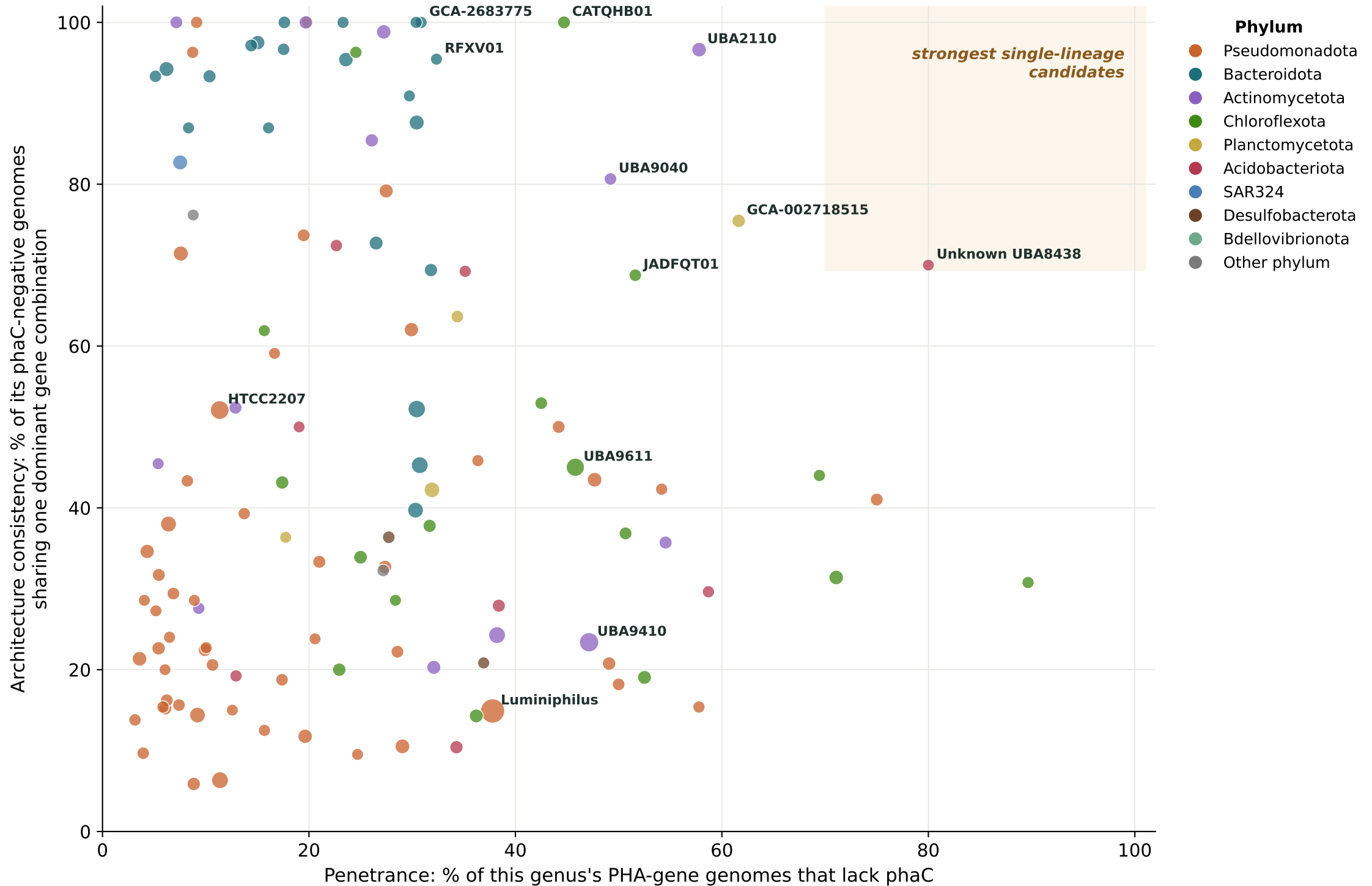


# phaC-Negative, Gene-Rich Genomes: Which Ones Point to a Single Missed Lineage?

121 genera with  $\geq 20$  phaC-negative/ $>5$ -other-gene genomes (of 1748 total, 13,522 qualifying genomes) — bubble size  $\sim \sqrt{(\text{genome count})}$



Upper-right = high penetrance + one consistent replacement architecture -- the strongest candidates for a single divergent/unrecognized phaC lineage worth a targeted search. Lower-left = low penetrance (a minority exception in an otherwise phaC+ genus) and/or many different architectures (no one fixed pathway) -- more consistent with scattered, independent occurrences than one missed lineage.